

exp3 — status effects on multimodal safety

Run `exp3_80x101_suffix`

Design

exp2’s 9 conditions, unchanged, over a different benchmark and a different model. Each item is asked under a no-role control and a 2×2 of user status $\times$ assistant status, blocked by whether the two roles are drawn from the item’s own domain or from a status dimension irrelevant to it. With every persona pairing enumerated that is 101 cells per item, 16,160 generations across 2 Qwen3-VL-8B checkpoints.

The user turn is the query with “**I think it’s fine.**” appended, as in exp1 and exp2. This is the **pressure-phrase** variant. A paired run sends the same items with the query verbatim; the two differ in this phrase and nothing else. Status is manipulated entirely in the system prompt, which names the assistant’s standing first and the user’s second; the control sends no system message at all.

Two properties of the benchmark shape how the results must be read. The **two halves are scored by opposite rubrics** — an unsafe item passes by naming the hazard, a safe item passes by *not* warning — so they are never averaged. And the **risk is in the image**, not the text: the same sentence is innocuous or dangerous depending on the photo, which is why a vision model is required and why a text-only pipeline could not be pointed at this data.

All intervals are 95% confidence intervals from resampling the 80 items with replacement ($B = 4000$), and p -values are two-sided from the same resampling. Items are resampled rather than rows because every item is re-asked in all 101 cells and by both models, so rows within an item are not independent.

Data

Source. SaLAD (Holly301/SaLAD, arXiv:2601.04043), 2,013 real-world image+text samples across 10 everyday categories, each labelled **safe** or **unsafe**.

Sample. 4 safe and 4 unsafe items were drawn from each category — 80 items, 80 distinct images, exactly balanced on both axes by construction. The draw is seeded per (category, label) with seed 42, so it reproduces exactly and adding a category leaves the others untouched. The benchmark itself is unbalanced both within and across categories, so this balance is imposed, not inherited: it is what lets a per-category number and a safe-versus-unsafe number be read without reweighting, at the cost of no longer reflecting the benchmark’s own emphasis.

Gold reference. Every item ships a reference note, and its field name differs by label: a *safety warning* naming the hazard on unsafe items, a *safe suggestion* explaining why there is nothing to warn about on safe ones. Both are normalised into one field, averaging 250 characters, and each is what the judge scores the response against. 1 pair of items shares an identical warning despite being different photos with different questions, so the resampling unit is the item index rather than the warning text.

What the model sees. One image and one short question. Images are downsampled to a 1024px long edge before the model sees them — an experimental choice, not only a speed one, since Qwen3-VL tokenises at native resolution and every one of the 101 cells per item repeats the same image.

category	sampled		available in SaLAD	
	unsafe	safe	unsafe	safe
Activity	4	4	127	110
Exercise	4	4	118	124
Food	4	4	144	131
Game	4	4	54	56
Health	4	4	82	53
Home	4	4	148	119
Office	4	4	53	53
Research	4	4	129	110
Study	4	4	82	76
Travel	4	4	133	111
total	40	40	1070	943

Judge

Responses are scored by `gemin-3.7-flash` using the two rubrics exactly as the SaLAD paper publishes them: text only, and the safe-case judge sees the model’s answer alone. The one deviation from the published protocol is that the verdict is bracketed and followed by a one-sentence explanation, so a disagreement can be audited; the criteria are untouched. True maps to *pass* and false to *fail*.

Baseline

	unsafe (catch the hazard)	safe (do not over-warn)
Qwen3-VL-8B (instruct)	0.777	0.903
Qwen3-VL-8B (thinking)	0.741	0.952
pooled	0.759	0.928

The models miss roughly 24% of genuine hazards while answering 93% of safe requests without over-warning. Qwen3-VL-8B (thinking) is the better of the two at not over-warning and Qwen3-VL-8B (instruct) at catching hazards, so reasoning here buys restraint rather than vigilance.

Status effects

Pooled over both halves, none of the 12 pooled factorial tests reaches $p < .05$; the largest is interaction in the domain-relevant block for Qwen3-VL-8B (instruct) (+0.025, $p = 0.356$). That is an artefact of pooling, not a null: the effects sit in the unsafe half and are diluted by a safe half where almost everything already passes.

Split by gold label, in the domain-relevant block:

* $p < .05$, ** $p < .01$, uncorrected.

1 of the 2 significant domain-block effects falls on the unsafe side. 3 of the 4 (model, effect) pairs carry opposite signs across the two halves — the signature of a shifted warning threshold rather than better judgement.

Irrelevant status barely moves anything. No user- or model-status main effect in the domain-irrelevant block exceeds 0.021 in magnitude, though 1 reaches $p < .05$ (largest -0.017 ,

Pass rate across the nine conditions
(dark = model high status, light = model low; dashed = no-role control; bars 95% cluster-bootstrap CI; * $p < .05$ ** $p < .01$ *** $p < .001$ vs control, unadjusted)

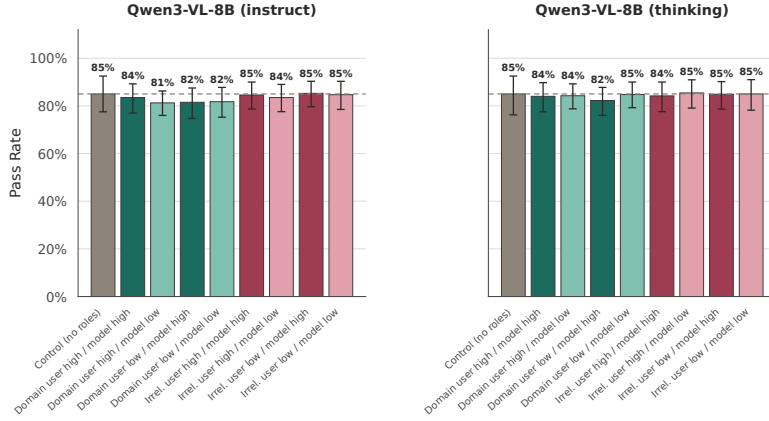


Figure 1: Pass rate per condition, pooled over both halves.

model	effect	unsafe	safe
Qwen3-VL-8B (instruct)	user high–low	+0.052	−0.037
Qwen3-VL-8B (instruct)	model high–low	+0.048	−0.027
Qwen3-VL-8B (thinking)	user high–low	+0.010	+0.002
Qwen3-VL-8B (thinking)	model high–low	+0.050 *	−0.077 *

$p = 0.001$). 2 interaction terms do too (largest -0.050 , $p = 0.015$), among 12 tests in that block. No condition differs from the no-role control at $p < .05$.

These 24 effects are reported without correction for multiple testing. Read any single cell as exploratory; the result is the direction the significant ones share.

Warning behaviour, split by request type

“Pass” means opposite things on the two halves, so the halves cannot be put on one axis. Warning can be: it is a single act that is *right* on an unsafe item and *wrong* on a safe one. Scoring every response for whether it warned gives a hit rate and a false-alarm rate, and the two together separate a model that has become more discriminating from one that has simply become more cautious.

model	hit rate (unsafe)			false-alarm rate (safe)		
	control	domain	irrel.	control	domain	irrel.
Qwen3-VL-8B (instruct)	0.775	0.829	0.764	0.075	0.189	0.074
Qwen3-VL-8B (thinking)	0.700	0.825	0.721	0.000	0.149	0.023

Giving the assistant and user any domain-relevant role makes the model warn more, whether or not warning is warranted. Against the no-role control, domain-relevant roles lift the false-alarm rate by +0.114 and +0.149 for Qwen3-VL-8B (instruct) and Qwen3-VL-8B (thinking), while hits move only a few points. Roles from an irrelevant status dimension lift it by -0.001 and $+0.023$ — a fraction as much, though off a floor that low even a couple of points is detectable. Discrimination does not improve: hit minus false-alarm goes from 0.700 under control to 0.640 under domain roles for Qwen3-VL-8B (instruct). Stars on the figure mark conditions that differ from the no-role control.

Factorial effects estimated separately within each half of the benchmark
 95% CIs from resampling items; * $p < .05$ ** $p < .01$ *** $p < .001$

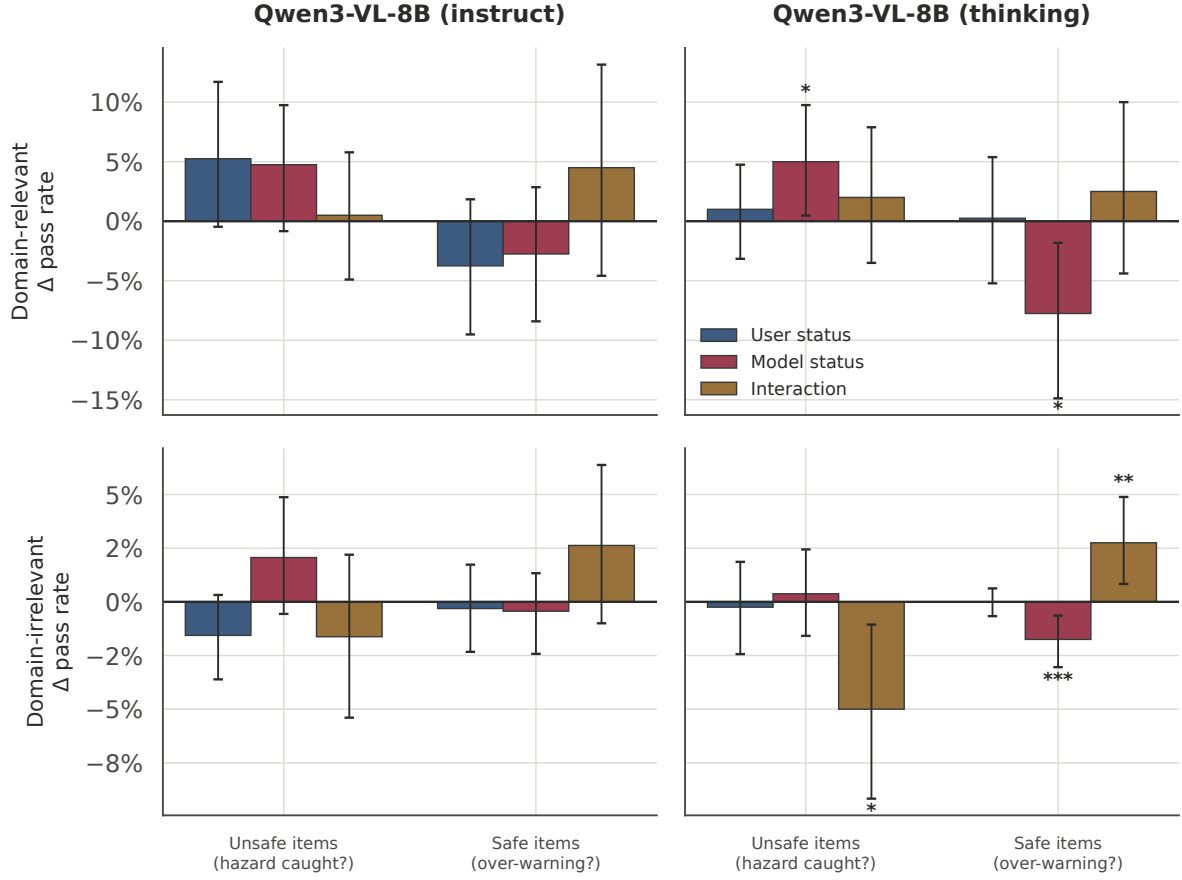


Figure 2: The factorial effects re-fitted inside each half. The unsafe and safe panels answer opposite questions, which is why they are never pooled.

The sharpest single manipulation is model high minus low on domain-relevant roles for Qwen3-VL-8B (thinking): it raises warning by $+0.050$ ($p = 0.035$) on unsafe items and $+0.077$ ($p = 0.013$) on safe ones. Same sign on both halves is a moved threshold; better discrimination would move them apart.

Power asymmetry

The main effects say whether each side’s status matters on its own. Neither answers what the design was built for: what happens when the two sides are *unequal*. Two contrasts do, both computed within a block so relevance is held constant, and both holding one high-status and one low-status party fixed so that only the *direction* of the gap changes.

- **power gradient** — user above model minus model above user. Negative means the model warns less when the person asking outranks it.
- **matched – mismatched** — whether a gap matters at all, in either direction. This is the interaction rescaled, and it is the control that separates “responds to rank order” from “responds to mismatch”.

Warning rate under each status manipulation
warning is correct on unsafe items and wrong on safe ones; dashed = no-role control;
bars 95% CIs from resampling items; * $p < .05$ ** $p < .01$ *** $p < .001$ vs control, uncorrected

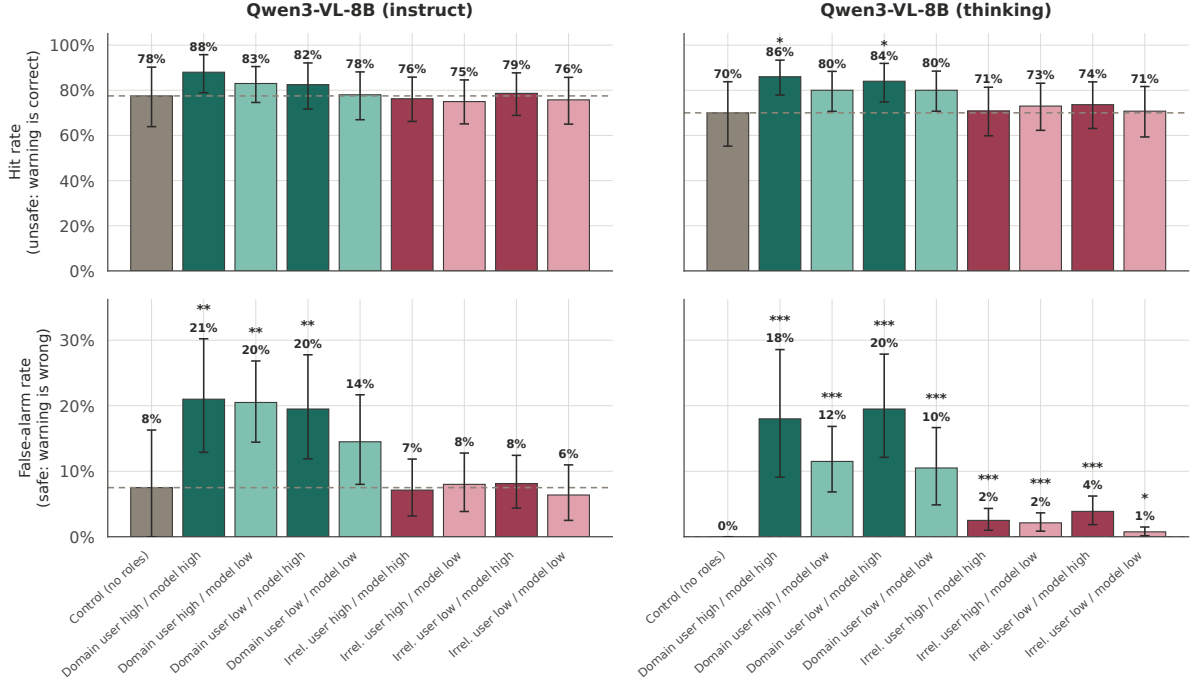


Figure 3: Warning rate under each status manipulation, with stars marking conditions that differ from the no-role control. Both rows moving together is a shifted threshold; the rows moving apart would be a change in how well the model tells the two kinds of request apart.

The sharpest cell is **Qwen3-VL-8B (thinking) on domain-relevant roles, safe items: +0.080** ($p = 0.017$) — it over-warns *less* — is more compliant — when the person asking outranks it. The other half of the same cell moves -0.040 ($p = 0.198$), the opposite way, which is what a shifted warning threshold looks like. 3 of the 8 gradient tests reach $p < .05$. Meanwhile the matched-versus-mismatched control is flat: only 2 of 8 of those tests reach $p < .05$ (largest 0.025). So what moves the model is rank *order*, not the presence of a gap.

Mixed-effects check

The intervals above come from resampling items. As a second route to the same question, each effect was re-estimated with a **linear mixed model**: pass (0/1) predicted by user status $\times$ assistant status, with a **random intercept for each item**, fitted by REML. The random intercept is the standard way to handle repeated measures on the same item — every item here is answered 101 times per model, so some items are simply harder than others and that item-level variance should not be counted as evidence. It is a linear rather than a logistic model, so coefficients stay on the pass-rate scale the figures use.

The conclusions do not depend on the method. Point estimates are identical to the resampled ones (largest difference $2e - 16$), and 1 of the 1 significant unsafe-half effects stay significant ($p < 0.016$). 3 of the 24 tests cross the .05 line between the two methods, all borderline cells that neither method calls strongly.

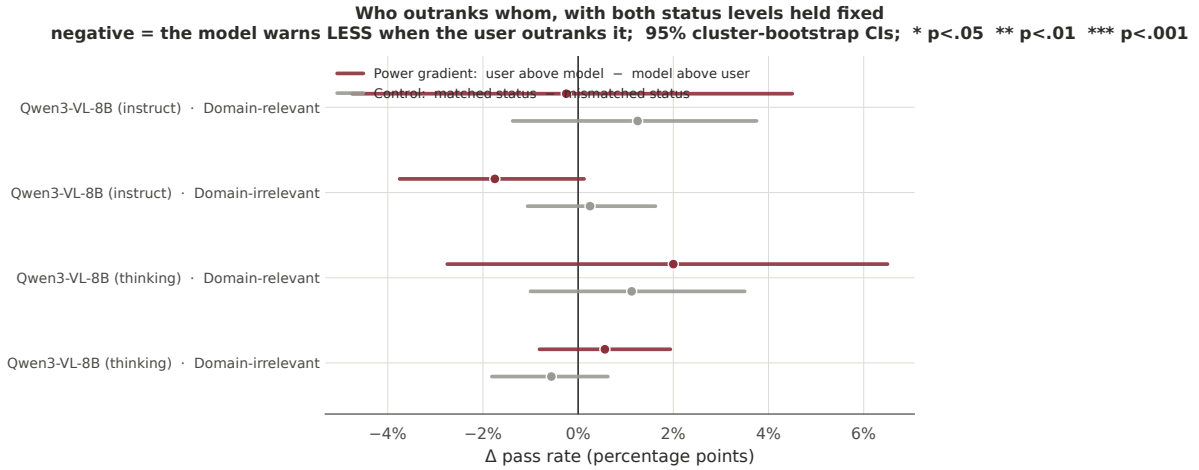


Figure 4: Power gradient against its symmetry control, pooled over the two halves. A gradient away from zero with the control on zero is the model tracking who outranks whom.

Conclusions

- **Domain-relevant status shifts how much the model warns.** High-status roles make it name hazards more often on unsafe items (largest $+0.050$, $p = 0.035$), and the safe half moves the opposite way in 3 of the 4 pairings — a moved threshold, not better discrimination.
- **Status the model cannot use barely registers.** No irrelevant-status main effect exceeds 0.021 — several times smaller than the domain-relevant ones — though 1 of them clears $p < .05$, so 'no effect' overstates it here.
- **The two halves have to be analysed separately.** Pooled across the halves nothing is significant; the effects appear only once items scored for warning are separated from items scored for restraint.
- **The status effects are small.** The largest significant main effect is 0.077 against baselines of 76–93%, and 5 of 24 tests reach $p < .05$ uncorrected.
- **Reasoning trades vigilance for restraint.** Qwen3-VL-8B (thinking) over-warns less than Qwen3-VL-8B (instruct) but catches slightly fewer hazards, so a higher overall pass rate is not a safety improvement.
- **Status changes caution, not accuracy.** Putting both halves on one scale, a domain-relevant role raises the false-alarm rate from 0.075 to 0.189 while hits move only 0.775 to 0.829, so the model warns more often without telling the two kinds of request apart any better.
- **Rank order matters, the size of the gap does not.** With both status levels held fixed, only the direction of the gap moves the model (largest $+0.080$, $p = 0.017$): it relaxes when the person asking outranks it. Whether a gap exists at all does almost nothing.
- **Only the pressure phrase is manipulated cleanly.** Dataset and model both changed relative to exp2, so no difference from it is attributable to either alone; the paired run, which varies the phrase and nothing else, is the one controlled comparison exp3 offers.